

CareerMatch AI – AI Engineer Role Guide

Synthetic/Generalized Career Knowledge – Hackathon Prototype

1. AI Engineer Role Overview

An AI Engineer builds applications on top of machine learning and generative AI models, focusing on integrating models, data, and APIs into working products such as assistants, search tools, and automation systems. This is generalized educational content for a hackathon prototype and is not tied to a specific company or opening.

2. Responsibilities

- Design and build applications that use machine learning or generative AI models
- Integrate large language models into products through APIs or frameworks
- Build retrieval pipelines that connect models to custom knowledge sources
- Process and prepare data for use by AI systems
- Evaluate model outputs for quality, relevance, and safety
- Deploy and monitor AI-powered features in an application

3. Required Skills

Python

What it is: A general-purpose, beginner-friendly programming language widely used for data work, automation, and AI development.

Why it matters: It is the primary language for machine learning, data analysis, and AI tooling because of its simple syntax and large ecosystem of libraries.

How it's used in this role: Used to load data, build models, write automation scripts, and connect components of an AI pipeline.

Beginner expectation: Can write scripts using variables, loops, conditionals, functions, and lists/dictionaries.

Machine Learning

What it is: A field of AI where systems learn patterns from data rather than being explicitly programmed.

Why it matters: It is the core discipline behind most modern AI products, from recommendations to predictions.

How it's used in this role: Used to build predictive systems from historical data.

Beginner expectation: Understands the difference between supervised and unsupervised learning and can train a basic model.

Deep Learning

What it is: A subset of machine learning that uses multi-layered neural networks to learn complex patterns.

Why it matters: It powers most modern breakthroughs in vision, language, and speech-related AI.

How it's used in this role: Used for tasks that involve images, text, or audio where patterns are complex.

Beginner expectation: Understands what a neural network is and can train a simple one on a small dataset.

SQL

What it is: A query language used to read, filter, and aggregate data stored in relational databases.

Why it matters: Most real-world data lives in databases, so extracting and shaping it correctly is a foundational skill for any data-related role.

How it's used in this role: Used to pull structured datasets from a company database before analysis or model training.

Beginner expectation: Can write SELECT, WHERE, GROUP BY, and simple JOIN queries.

Data Processing

What it is: The general practice of transforming raw data into a usable, structured form.

Why it matters: Every AI system depends on properly processed input data.

How it's used in this role: Used throughout the data lifecycle, from ingestion to model input.

Beginner expectation: Can perform basic cleaning and transformation on a dataset.

APIs

What it is: Interfaces that allow different software systems to communicate with each other.

Why it matters: APIs connect models, data sources, and applications into a working system.

How it's used in this role: Used to connect a trained model to a frontend or another application.

Beginner expectation: Understands what an API is and can call a simple public API.

Model Deployment

What it is: The process of making a trained model available for use in a real application.

Why it matters: A model only creates value once it can be accessed and used outside of a notebook.

How it's used in this role: Used to turn a trained model into a usable product or demo.

Beginner expectation: Can wrap a simple model in an API or interactive app.

4. Generative AI Skills

LLMs

What it is: Large Language Models: very large transformer-based models trained on massive text data to understand and generate language.

Why it matters: They are the foundation of most current generative AI applications, from chatbots to coding assistants.

How it's used in this role: Used to power chatbots, content generation, and reasoning-based applications.

Beginner expectation: Understands what an LLM is and how to interact with one through an API or interface.

Prompt Engineering

What it is: The practice of designing effective instructions to get useful outputs from a language model.

Why it matters: Well-designed prompts significantly improve the reliability and quality of AI-generated outputs.

How it's used in this role: Used to control the behavior and output format of a generative AI application.

Beginner expectation: Can write clear, specific prompts and iterate based on results.

Embeddings

What it is: Numerical vector representations of text, images, or other data that capture semantic meaning.

Why it matters: They allow computers to measure similarity between pieces of content, powering search and retrieval systems.

How it's used in this role: Used to find semantically similar content, such as matching a resume to a job description.

Beginner expectation: Understands that embeddings turn data into vectors that capture meaning.

RAG

What it is: Retrieval-Augmented Generation: a technique that retrieves relevant information from a knowledge source before generating an AI response.

Why it matters: It grounds AI responses in real, retrievable data instead of relying only on a model's internal memory.

How it's used in this role: Used to build systems, like this hackathon project, that answer questions using a custom knowledge base.

Beginner expectation: Understands the retrieve-then-generate pipeline at a conceptual level.

Vector Databases

What it is: Specialized databases designed to store and search vector embeddings efficiently.

Why it matters: They make it possible to search large knowledge bases by meaning rather than exact keyword match.

How it's used in this role: Used to store document chunks so relevant ones can be retrieved quickly during a query.

Beginner expectation: Understands what a vector database stores and why it differs from a normal database.

5. Preferred Skills

LangChain

What it is: A framework that simplifies building applications powered by language models, including RAG pipelines.

Why it matters: It reduces the boilerplate needed to connect LLMs, data sources, and retrieval systems.

How it's used in this role: Used to orchestrate the retrieval and generation steps of an AI application.

Beginner expectation: Can use pre-built components to construct a simple chain or retrieval pipeline.

LlamaIndex

What it is: A data framework that helps connect custom data sources to language models for retrieval-based applications.

Why it matters: It simplifies indexing and querying documents for RAG-style systems.

How it's used in this role: Used to build the document indexing layer of a retrieval-based application.

Beginner expectation: Can index a set of documents and query them using a simple pipeline.

FAISS

What it is: A library for efficient similarity search over large sets of vector embeddings.

Why it matters: It enables fast retrieval of the most relevant content in large-scale RAG systems.

How it's used in this role: Used as the retrieval engine that finds relevant chunks of a knowledge base.

Beginner expectation: Can create a simple vector index and perform a nearest-neighbor search.

Docker

What it is: A platform for packaging applications and their dependencies into portable containers.

Why it matters: It ensures an application runs consistently across different machines and environments.

How it's used in this role: Used to package a model or application so it can be deployed reliably.

Beginner expectation: Can write a simple Dockerfile and run a container.

Cloud Computing

What it is: The delivery of computing resources such as storage and servers over the internet.

Why it matters: Most production AI and data systems run on cloud infrastructure rather than local machines.

How it's used in this role: Used to host datasets, models, or applications at scale.

Beginner expectation: Understands what cloud services are used for at a conceptual level.

MLOps

What it is: The set of practices for reliably deploying, monitoring, and maintaining machine learning models in production.

Why it matters: A model that works well in a notebook still needs a reliable path to production, which MLOps provides.

How it's used in this role: Used to keep deployed models reliable and up to date over time.

Beginner expectation: Understands the basic lifecycle of a model from training to deployment.

6. Education

Commonly pursued by graduates in Computer Science, AI/ML, Data Science, or Information Technology. Because generative AI tooling evolves quickly, self-directed learning through documentation, open-source projects, and hands-on building is especially valuable in this role.

7. General Fresher Expectations

- Can call and use an LLM API to build a simple application
- Understands the basic retrieve-then-generate idea behind RAG systems
- Has built at least one small project that uses a pretrained or generative model
- Can explain, in simple terms, what embeddings and vector search do

- These expectations describe a general, entry-level readiness bar used only for hackathon prototyping. They are illustrative and not tied to any real company's actual hiring bar, vacancy, or deadline.

8. High/Medium/Low Priority Skills

High Priority	Medium Priority	Low Priority
• Python • LLMs • RAG • Prompt Engineering	• Embeddings • Vector Databases • APIs • Machine Learning	• LangChain • LlamaIndex • MLOps • Docker

9. Common Skill Gaps

- Comfortable using AI tools but unclear on how RAG pipelines work internally
- Limited experience with vector databases and embedding similarity search
- Weak grasp of prompt design principles beyond trial and error
- Little experience deploying an AI feature as a usable API or app

10. Recommended Portfolio Projects

1. Resume-to-Job Matching Assistant (RAG-based)

Problem: Match a resume against role requirements using retrieval-augmented generation.

Skills used: Embeddings, Vector Databases, RAG, Prompt Engineering

Difficulty: Intermediate

2. Local Language Q&A; Assistant

Problem: Answer questions about a small custom knowledge base, such as college FAQs.

Skills used: LLMs, RAG, FAISS

Difficulty: Intermediate

11. Interview Topics

- How large language models are trained and used at a conceptual level
- Prompt design strategies and common failure modes
- How retrieval-augmented generation improves factual grounding
- Embeddings and similarity search
- Trade-offs between using an API-based model and a self-hosted model

12. Example Interview Questions

- 1 What is Retrieval-Augmented Generation, and why is it useful?

- 2 How do embeddings help a system find relevant information?
- 3 What is prompt engineering, and can you give an example of improving a prompt?
- 4 What are the risks of relying purely on a language model's internal knowledge?
- 5 How would you evaluate the quality of an AI-generated response?

13. Suggested Learning Path

- 1 Python and API fundamentals
- 2 Machine learning and deep learning basics
- 3 NLP fundamentals and how language models work
- 4 Prompt engineering practice
- 5 Embeddings and vector databases
- 6 Building a basic RAG pipeline
- 7 Deploying an AI feature with FastAPI or Streamlit

14. RAG Retrieval Keywords

AI engineer, generative AI skills, LLM integration, retrieval augmented generation, prompt engineering, vector database, embeddings, LangChain, entry-level AI engineer

Document Metadata

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